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Design of Flash ADC using low offset comparator for analog signal processing application

Anurag Yadav[1], Nivedita Rai[2], Aishita Verma[3], Subodh Wairya[4]

Department of Electronics and Communication Engineering,
Institute of Engineering and Technology,
Lucknow, India
anuraglko2014@gmail.com[1], niveditarai95@gmail.com[2], aishita.verma19@gmail.com[3], swairya@gmail.com[4]

***Abstract*—The analog to digital converters (ADCs) are used in high bandwidth applications such as communications, radar processing and data acquisition system. In this work, a 3-bit Flash ADC is designed on Cadence simulation software with 180nm CMOS technology. The important parameters of ADC are resolution, conversion time and power dissipation. Flash ADC converts analog input signal to digital output signal in one clock cycle. In this work, Flash ADC has been designed using two-stage operational amplifier as comparator, voltage divider circuit and priority encoder. The proposed prototype has been simulated with 2.8V supply voltage at 27°C. The simulation results for the design shows an overall power dissipation of 3.6947mW for flash ADC, 431.637µW for the comparator. The main advantage of the comparator used is its low input offset voltage of 7.84 mV. The gain of the two stage operational amplifier is 85dB with the Miller Compensation Capacitance of 800fF and load capacitance of 1pF. This ADC has high input signal voltage range which is 0 to 2.8V. Monte Carlo simulation is done for power dissipation with 300 samples. Process variation is done for the power dissipation at different corners.**

***Keywords*—Flash ADC, Two-stage operational amplifier, Comparator, Priority Encoder, Monte Carlo simulation, Process variation.**

I. INTRODUCTION

For conversion from analog to digital there are different types of ADCs [1]. Flash ADC is simplest in terms of its operation and is the most efficient ADC technologies in terms of speed. It provides an interface between real and digital world. ADCs have wide range of applications. An ADC must be of low latency in such a case flash ADC is the ideal choice in terms of speed as it is the fastest ADC[2].The flash ADC technique does not require any linear amplification as it has highest rate of conversion. The conversion speed of a flash ADC is very fast as compared to the other ADCs. Flash ADC contains an array of comparators, e.g. for 3-bit flash ADC 7 comparator's are required. Flash ADC only used in applications where (a) latency is paramount (b) hardware complexity is modest. Every time resolution is increased by 1-bit, the number of comparators doubles so hardware complexity increases exponentially as the number of bits.

Comparator bank output is applied to the input of priority encoder [3] logic circuit that produces unique digital code for each range of the voltage [4]. The priority encoder is designed for achieving the ultra low power consumption. The comparator has been implemented by two-stage operational amplifier as it provides high gain [5]. Figure 1 depicts the structure of a typical flash ADC [6]. Due to contemporaneous conversion of the signal from continuous time signal to digital signal, the Flash ADC has substantiated to be fastest.

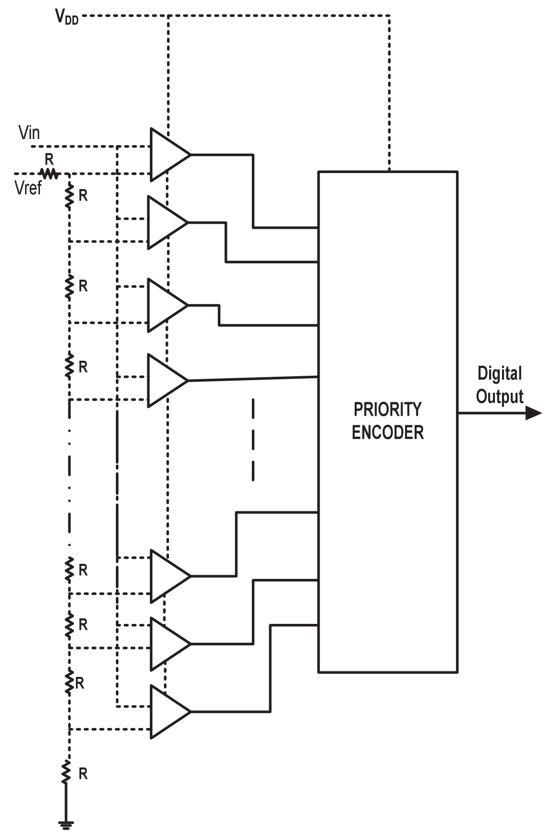

Fig. 1. Block diagram of conventional flash ADC [3]

The paper is organized as follows: Section II depicts the design of proposed flash ADC using voltage divider, comparators and a priority encoder. Section III depicts the design of comparator using two stage opamp, it has input offset voltage of 7.84 mV and gain of 85dB. Section IV depicts the design of priority encoder. Section V, performs a simulation setup result and performance analysis result for proposed flash ADC in terms of Monte Carlo for 300 samples, process variation is done for the power dissipation at different corners. The last section concludes this paper. All the simulations has been carried out in a cadence 180nm CMOS technology.

II. PROPOSED FLASH ADC

The proposed flash ADC consist of three major components High speed comparator, resistive voltage divider network and priority encoder in its structure. Figure 2 illustrates the schematic of Proposed 3 Bit Flash ADC. A 3-bit Flash ADC consists of 8 resistors which are providing voltage divider network [7-9].This voltage divider network is resulting into fixed voltages which are been given to comparator [10]. The voltages of voltage divider network is connected with negative terminals of comparator and the second terminal of input that is positive terminal is connected to the input signal [11]. The output of priority encoder is based on the priority given at the input of priority encoder [12]. The conversion of analog to digital data happens by just single cycle, which concludes that it is fastest flash ADC.

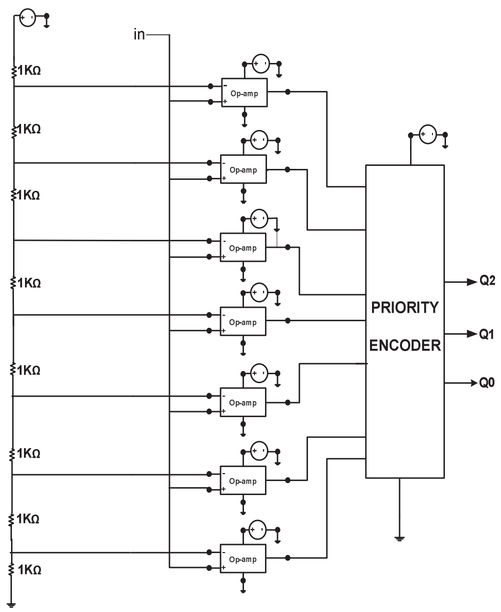

Fig. 2. Schematic of 3 Bit Flash ADC

III. COMPARATOR DESIGN

A. Two stage opamp as Comparator Design

Operational amplifier (op-amp) has many applications in analog circuits where one of the important applications is comparator. Comparator is the most important analog building block of any Flash ADC [16-17]. A comparator, as its name implies does a comparison task between two applied inputs and the dominant one amongst the two inputs decides the output of the comparator [18]. In Figure 3 an inverting input or negative terminal is supplied with reference voltage whereas a non-inverting terminal or positive terminal fed with input voltage. This comparator acts like a one bit of a analog to digital converter[19]. Comparator has a open collector configuration which make it convenient to use in different logic families.

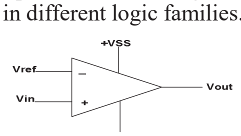

Fig. 3. Basic comparator symbol [8]

The conditions for figure 3 is explained in the (1) and (2) below

If Vin > Vref then V_{out} = logic 1 (1)
If Vin < Vref then V_{out} = logic 0 (2)

In the design of an operational amplifier (op-amp), differential amplifier is the initial building block[13] with some small gain enhanced by the amplifier with active loads. This paper carries out the simulation of the two-stage operational amplifier performed on 180nm technology. Supply voltage and channel length are the salient parameters in any op-amp design. NMOS differential gain stage followed by a PMOS connected common source (CS) amplifier (M5) are two levels together coining the two-stage of the operational amplifier [14].The advantage of using PMOS connected CS is that it provides large voltage swing and large gain as they are active loads. The differential gain stage is a single pole system consisting of identical structure where aspect ratio or size of the transistor M1, M2 and M3, M4 is same. Transistors M3 and M4 forms the current mirror circuit that acts a active load device resulting in a very large output resistance. The bias voltage and size of the transistor M7 should be selected such that drain current is sum of the two identical currents flowing in two branches of the differential gain [15] and also the output resistance of M7 is coming out large as shown in figure 4.

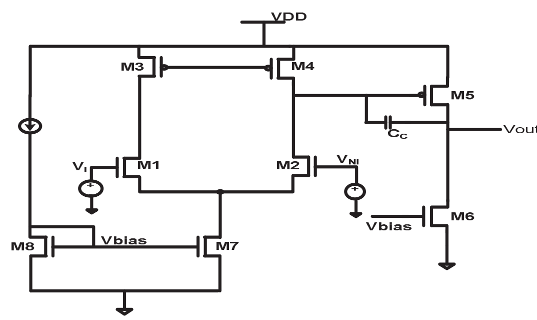

Fig. 4. Schematic of comparator (two stage op-amp design with 1pF load) [9]

To fulfill the requirement of high gain and large swing, second stage is added to differential gain stage. This stage consists of common source (CS) amplifier as it stacks less number of transistors and also helps to boost the gain[20]. Here, transistor M5 is a PMOS connected common source stage with current source M6 biased through gate to source voltage (Vgs) of the current mirror load M7 and M8.The gain of the differential stage amplifier is given in (3)

$$gain1 = gm2(ro2||ro4) = gm2/(gds2+gdS4) \quad (3)$$

Where gm_2 is the transconductance of transistor M_2 and drain to source transconductance of M_2 and M_4 are represented by gds_2 and gds_4.The gain of CS amplifier is given by (4)

$$gain2 = gm5(ro6||ro5) = gm5/(gds6+gdS5) \quad (4)$$

Where gm6 is the transconductance of transistor M_5 and drain to source transconductance of M_5 and M_6 are represented by gds_5 and gds_6.

Total gain is given by (5)

$$\text{gain} = \text{gain1} + \text{gain2} \tag{5}$$

where gain1 and gain2 are calculated in decibels(dB).
The open loop gain of op-amp is calculated as product of each gain stage as given in (6)

$$A = A1 \times A2 = gm1(r01 \| r04) \times gm2(r06 \| r07) \tag{6}$$

A_1 is gain generated by the fully differential amplifier, while A_2 is the gain for the PMOS connected common source amplifier, gm1 and gm2 are the transconductance of first and second stage amplifier respectively and r01, r04, r06 and r07 are the output resistance of transistors M_1, M_4, M_6 and M_7.

B. Compensating the op-amp design

Compensation is a method by which op-amps are stabilized through Miller capacitor C_C connected in a negative feedback fashion between output of differential gain stage and CS stage[20-21]. Two-stage op-amp is a two pole system as we have two RC circuits in its small signal model analysis. Miller capacitor acts a pole-splitter. During bode plot analysis, it is found that pole of the CS amplifier (P_2) gives approx. zero degree phase margin which causes ringing effect in the output. A phase margin is a measure of stability in the system. More the phase margin of the system, more will be the stability and less ringing effect will occur in the system [22].To remove the ringing effect in two stage op-amp, the dominant pole concept is implemented which effectively makes the system as a one pole system even though it consists of two poles. Since the second stage capacitance consists of load capacitance of 1pf which is difficult to change so it's a challenge to move pole P_2 so movement of pole of the first stage amplifier (P_1) is done by increasing the value of first stage capacitance. Here pole P_1 is moved in such a way that gain crosses 0 dB before pole P_2 comes, so this makes pole P_2 ineffective [23].

Requirement for minimum value of C_C is given by (7)

$$C_C \geqslant 0.22 C_L \tag{7}$$

So, the minimum miller capacitance taken here is 800 fF.
Gain bandwidth (GBW) product is given by (8)

$$GBW = gm1/C_C \tag{8}$$

Where gm1 is the transconductance of differential gain stage and C_C is the compensation capacitor (Miller compensation).

IV. Design of the Priority Encoder

Priority encoder is the combinational circuit which has 2^N inputs and N output. The output is corresponding to the input with highest preference. The unique attribute of priority encoder is that it overcomes the setback by the basic encoder circuit. Therefore, mostly priority encoders are given preferences over the basic encoder in the design of flash ADC circuit [24]. The priority encoder as the name suggests, assigns a priority level to each input according to its position weighting and output corresponds to that input having the highest priority. The block diagram of Priority encoder is shown in figure 5 and table I is the truth table of Priority encoder. Table I shows the conversion of analog signal (V_{in}) to the digital at the output of the priority encoder.

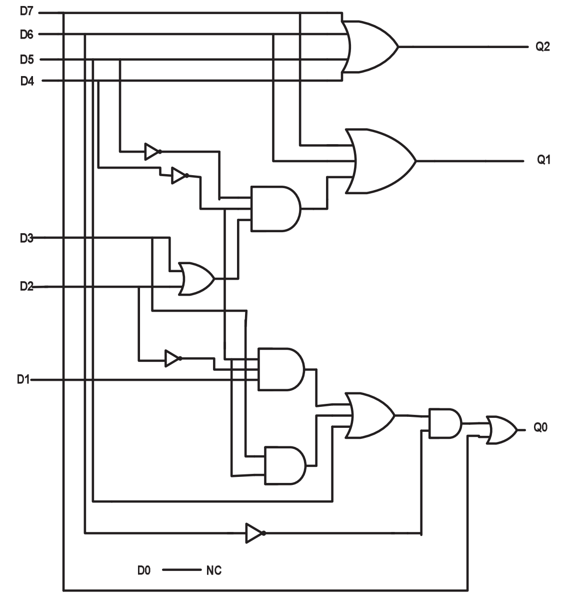

Fig. 5. Block diagram of Priority Encoder using Logic Gates [5]

TABLE I. Truth table of Priority Encoder corresponding to input signal range

Input Signal Range Vin (V)	Priority Encoder Output		
	Q2	*Q1*	*Q0*
0<Vin<0.35	0	0	0
0.35<Vin<0.7	0	0	1
0.7<Vin<1.05	0	1	0
1.05<Vin<1.4	0	1	1
1.4<Vin<1.75	1	0	0
1.75<Vin<2.1	1	0	1
2.1<Vin<2.45	1	1	0
2.45<Vin<2.8	1	1	1

V. Simulation Results and Analysis

A. Priority Encoder

In the priority encoder if more than one input is active, the one with highest order decimal digit will be active; the higher input is encoded at the output [14]. In the truth table I, there are eight inputs and three outputs in the truth table I and all the inputs and outputs are included in the truth table I. All the input combinations are verified with the help of truth table. The simulation waveform of priority encoder is shown in figure 6.

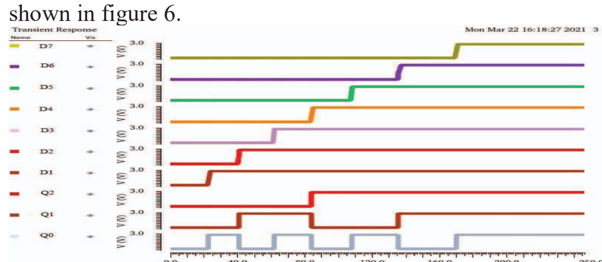

Fig. 6. Simulation waveform for Priority Encoder

B. Comparator

The complete comparator design (two stage op-amp circuit) is shown in Figure 4 with the nominal device sizing shown in table II. All the devices are maintained in the saturation region with the help (9) to get large gain and high voltage swings. A Miller capacitor C_C of 800 fF is used for stabilizing the op-amps.

$$Vds \geq Vgs\text{-}Vt \tag{9}$$

Where Vds is drain to source voltage, Vgs is gate to source voltage and Vt is threshold voltage. A DC analysis is done to get the power of 431.637µW at load capacitance C_L of 1pF at 27°C with common mode voltage of 1.4V. The gain obtained is 85dB. Table II illustrates about the aspect ratio of the MOS transistor in µm followed by table III which shows the parameters of two stage op-amp. The plot of gain (dB) and phase (degree) of two stage op-amp is given in figure 7.

TABLE II. TRANSISTOR SIZES FOR COMPARATOR (OP-AMP DESIGN)

MOS Transistor	W/L (µm)
M1,M2	6/1
M3, M4	14/1
M5	174/1
M6	75/1
M7, M8	12/1

TABLE III. SIMULATION RESULTS FOR COMPARATOR (OP AMP DESIGN)

Comparator (Op-Amp design) Parameters	Value
Gain	85dB
Phase	55.69 °
Unity gain Bandwidth	22.377MHz
3-dB Frequency	1.512KHz
Miller Compensation Capacitance (C_C)	800fF
Input referred noise (V^2/Hz)	1.31807×10^{-13}
Input Offset Voltage	7.84 mV

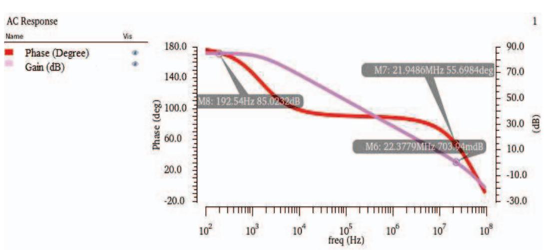

Fig. 7. Plot of Gain and Phase margin of two stage op-amp

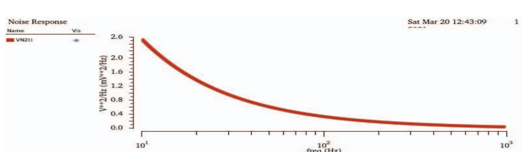

Fig. 8. Noise Analysis of comparator (Two-Stage Op-amp)

The Figure 8 shows the noise analysis of the Two-Stage Op-amp in the frequency range of 10Hz to 1KHz thereby obtaining total input referred noise of $1.31807\times10^{-13}\,V^2$/Hz. The input referred noise is the noise voltage when it applied to the input of the noiseless circuit it create same amount of noise at the output as the real circuit does. Figure 9 depicts input offset voltage for comparator. The difference between the output and reference voltage along x- axis is calculated as offset voltage, which is 7.84 mV.

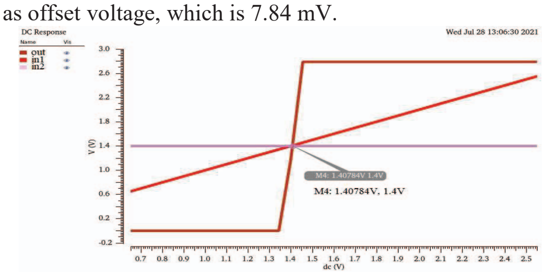

Fig. 9. Input Offset Voltage for comparator design

Two stage opamp is used as a comparator in this design. The main advantage of this comparator is its low input offset voltage of 7.84 mV. To get better performance of two stage opamp as a comparator remove the miller compensation capacitor C_C, it increases the value of first pole frequency of opamp. It increases the effectiveness of comparator in taking the decision of logic high and logic low for the flash ADC in lower reference voltages. Table IV depicts the input offset voltage comparison for designed and existing comparators.

TABLE IV. INPUT OFFSET VOLTAGE COMPARISON FOR COMPARATORS

Parameter	This Work	[27]	[26]	[25]
Supply Voltage (V)	2.8	5	2.5	3.3
Input Offset Voltage (mV)	7.84	8.4	20	50
Technology (nm)	180	180	180	350

C. Flash ADC output

The complete structure of 3 bit flash is shown in the figure 2 and with the three components high speed comparator, resistive voltage divider network and priority encoder in its structure. The simulation waveform of flash ADC is shown in Figure 10. Flash ADC is simulated in the cadence environment using 0.18µm technology and the total power consumed by 3 bit flash ADC circuit is 3.6947mW. The average transient power is 2.3mW and DC power is 1.393mW. The Table V illustrates the simulation specifications of the proposed Flash ADC.

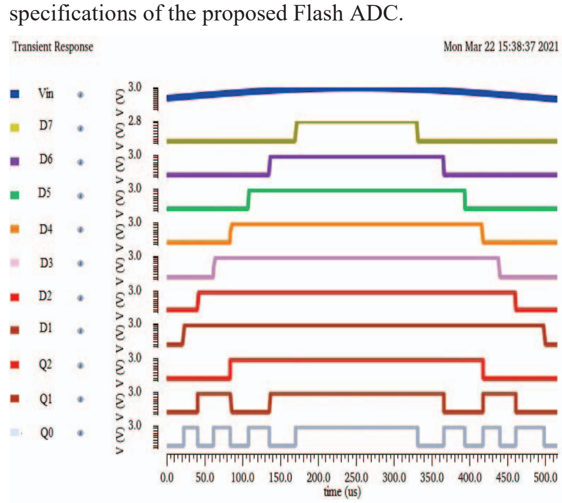

Fig. 10. Simulation waveform (transient analysis) for Flash ADC

TABLE V. SIMULATION SPECIFICATIONS OF PROPOSED FLASH ADC

Parameters	Proposed Flash ADC
Technology	180nm
Resolution	3bit
Supply Voltage	2.8V
Input Signal Voltage Range	0V-2.8V
Power Dissipation Of Encoder	1.58mW
Power Dissipation Of Comparator	431.637µW
Total Power Dissipation Of Flash ADC	3.6947mW

D. *Monte Carlo Simulation*

This simulation is done for 300 samples in which the sigma value is taken as 3 for avg. dynamic power dissipation and static power dissipation. In the average dynamic power dissipation, the mean value is found 2.309mW and standard deviation is 41.99µW. Similarly, for the static power dissipation the mean value is 1.394mW and standard deviation is 12.94µW. In both the cases deviation of samples are very close to mean. This shows the performance of the circuit is good due to less deviation of samples from their mean value. The simulations are shown in figure 11 and figure 12. Also figure 13 shows simulation for total power consumption. Table VI shows the Monte Carlo simulation results for power dissipation.

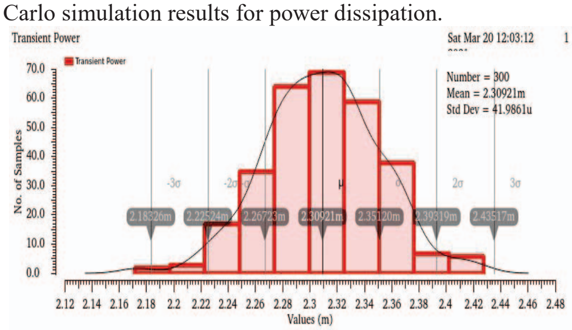

Fig. 11. Monte Carlo simulation for average dynamic power dissipation

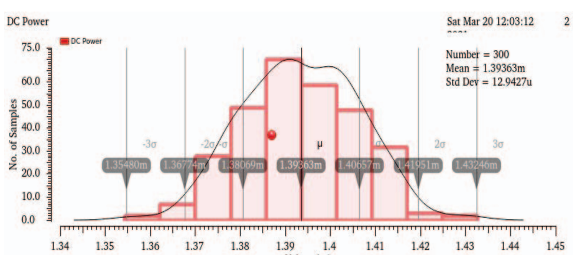

Fig. 12. Monte Carlo simulation for static power dissipation

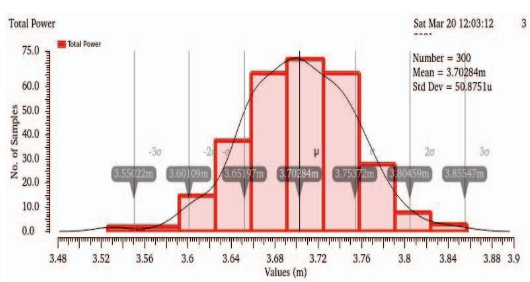

Fig. 13. Monte Carlo Simulation for Total power consumption

TABLE VI. MONTE CARLO SIMULATION RESULTS

Output	Mean	Median	Std. Dev.
Average Dynamic power dissipation	2.309 mW	2.308 mW	41.99 µW
Static power dissipation	1.394 mW	1.393 mW	12.94 µW
Total power dissipation	3.703 mW	3.704 mW	50.88 µW

E. *Process variation*

Corner analysis is done in four different conditions of MOS transistors. These are FF, FS, SF, SS and nominal. Table VII depicts the value of power dissipation for circuit under different condition of MOS transistors. Total power dissipation of flash ADC for SF corner is more and for Nominal, it is less. Corner analysis for total power consumption of Flash ADC is shown in figure 14.

TABLE VII. CORNER ANALYSIS FOR FLASH ADC

Parameters	SS	FF	FS	SF	Nominal
Average Dynamic power dissipation (mW)	2.478	2.309	2.307	2.492	2.307
Static power dissipation (mW)	1.378	1.388	1.388	1.378	1.387
Total power dissipation (mW)	3.856	3.697	3.694	3.87	3.694

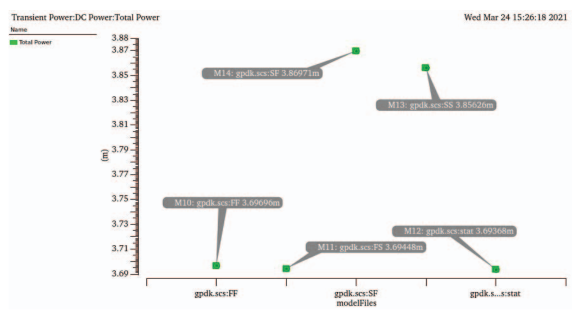

Fig.14. Corner analysis for Total Power Consumption of Flash ADC

VI. CONCLUSION

The proposed Flash ADC with priority encoder and comparator consumes less power. The AC, DC and transient simulation are carried out in cadence environment using virtuoso tool under 180nm technology. The simulation results for the design illustrate power dissipation of 431.637µW for the comparator and 3.6947mW for 3-bit Flash ADC with 2.8V supply. The average transient power is 2.3mW and DC power is 1.393mW for the 3-bit flash ADC. The important block of this flash ADC is comparator. Two stage opamp is used as a comparator in this design. It has low input offset voltage of 7.84 mV. There is 6.67% reduction in the input offset voltage when compared with [27] and 84.32 % reduction when compared with [25]. The gain obtained for the two-stage op-amp is 85dB using Miller Compensation Capacitance (C_C) of 800fF and load capacitance (C_L) of 1pF at the common mode input voltage of 1.4V. The total input referred noise of the two stage op-amp is 1.31807×10^{-13} V^2 /Hz. The advantage of this flash

ADC is that it can convert the analog input signal within the range of 0 to 2.8V. Corner analysis shows the 4.76% maximum increase in the total power dissipation when compared with the nominal case.

REFERENCES

[1] Jan M Rabaey, Anantha Chandrakasan, Borivoje Nikolic, "Digital Integrated Circuits, a design perspective", second edition, Prentice Hall 2011.

[2] Jincheol Yoo, Kyusun Choi and A. Tangel, "A 1-GSPS CMOS flash A/D converter for system-on-chip applications," Proceedings IEEE Computer Society Workshop on VLSI 2001. Emerging Technologies for VLSI Systems, Orlando, FL, USA, 2001, pp. 135-139, doi: 10.1109/IWV.2001.923152.

[3] A. Chunn and R. K. Sarin, "Comparison of thermometer to binary encoders for flash ADCs," 2013 Annual IEEE India Conference (INDICON), Mumbai, 2013, pp. 1-4, doi: 10.1109/INDCON.2013.6726138.

[4] V. Shukla, O. P. Singh, G. R. Mishra and R. K. Tiwari, "A novel approach to design decimal to BCD encoder with reversible logic," 2014 International Conference on Power, Control and Embedded Systems (ICPCES), Allahabad, 2014, pp. 1-5, doi: 10.1109/ICPCES.2014.7062823.

[5] Tsung-Sum Lee, Li-Dyi Luo and Chin-Sheng Lin, "Design techniques for a CMOS low-power low-voltage fully differential flash analog-to-digital converter," The 2004 47th Midwest Symposium on Circuits and Systems, 2004. MWSCAS '04., Hiroshima, Japan, 2004, pp. I-357, doi: 10.1109/MWSCAS.2004.1354001.

[6] S. Veeramachanen, A. M. Kumar, V. Tummala and M. B. Srinivas, "Design of a Low Power, Variable-Resolution Flash ADC," 2009 22nd International Conference on VLSI Design, New Delhi, 2009, pp. 117-122, doi: 10.1109/VLSI.Design.2009.62.

[7] C. Wu, C. Chang and M. Ho, "A subdivision method for improving resolution of analog encoders," 2017 International Automatic Control Conference (CACS), Pingtung, 2017, pp. 1-6, doi: 10.1109/CACS.2017.8284255.

[8] C. Vudadha et al., "Low-Power Self Reconfigurable Multiplexer Based Decoder for Adaptive Resolution Flash ADCs," 2012 25th International Conference on VLSI Design, Hyderabad, 2012, pp. 280-285, doi: 10.1109/VLSID.2012.84.

[9] Zhaohui Huang and Peixin Zhong, "An adaptive analog-to-digital converter based on low-power dynamic latch comparator," 2005 IEEE International Conference on Electro Information Technology, Lincoln, NE, 2005, pp. 6 pp.-6, doi: 10.1109/EIT.2005.1627029.

[10] A. Khorami, M. B. Dastjerdi and A. F. Ahmadi, "A low-power high-speed comparator for analog to digital converters," 2016 IEEE International Symposium on Circuits and Systems (ISCAS), Montreal, QC, 2016, pp. 2010-2013, doi: 10.1109/ISCAS.2016.7538971.

[11] Y. Gupta, L. Garg, S. Khandelwal, S. Gupta and S. Saini, "Design of low power and high speed multiplexer based Thermometer to Gray encoder," 2013 International Symposium on Intelligent Signal Processing and Communication Systems, Naha, 2013, pp. 501-504, doi: 10.1109/ISPACS.2013.6704602.

[12] M. Nasrollahpour, C. Yen and S. Hamedi-hagh, "A high-speed, low-offset and low-power differential comparator for analog to digital converters," 2017 International SoC Design Conference (ISOCC), Seoul, 2017, pp. 220-221, doi: 10.1109/ISOCC.2017.8368862.

[13] C. Chen, J. Wu, J. Hung, T. Li, W. Liu and W. Shih, "A 12-Bit 3 GS/s Pipeline ADC With 0.4 mm^{2} and 500 mW in 40 nm Digital CMOS," in IEEE Journal of Solid-State Circuits, vol. 47, no. 4, pp. 1013-1021, April 2012, doi: 10.1109/JSSC.2012.2185192.

[14] K. Mei, N. Zheng, C. Huang, Y. Liu and Q. Zeng, "VLSI Design of a High-Speed and Area-Efficient JPEG2000 Encoder," in IEEE Transactions on Circuits and Systems for Video Technology, vol. 17, no. 8, pp. 1065-1078, Aug. 2007, doi: 10.1109/TCSVT.2007.903555.

[15] S. Veeramachanen, A. M. Kumar, V. Tummala and M. B. Srinivas, "Design of a Low Power, Variable-Resolution Flash ADC," 2009 22nd International Conference on VLSI Design, New Delhi, 2009, pp. 117-122, doi: 10.1109/VLSI.Design.2009.62.

[16] S. Nahata, Kyusun Choi and Jincheol Yoo, "A high-speed power and resolution adaptive flash analog-to-digital converter," IEEE International SOC Conference, 2004. Proceedings., Santa Clara, CA, USA, 2004, pp. 33-36, doi: 10.1109/SOCC.2004.1362342.

[17] M. Moslemi and S. Babayan-Mashhadi, "A novel power-efficient architecture for high-speed flash ADCs," 2014 22nd Iranian Conference on Electrical Engineering (ICEE), Tehran, 2014, pp. 247-250, doi: 10.1109/IranianCEE.2014.6999541.

[18] Yao-Jen Chuang, Hsin-Hung Ou and Bin-Da Liu, "A novel bubble tolerant thermometer-to-binary encoder for flash A/D converter," 2005 IEEE VLSI-TSA International Symposium on VLSI Design, Automation and Test, 2005. (VLSI-TSA-DAT)., Hsinchu, Taiwan, 2005, pp. 315-318, doi: 10.1109/VDAT.2005.1500084.

[19] G. T. Varghese and K. K. Mahapatra, "A high speed low power encoder for a 5 bit flash ADC," 2012 International Conference on Green Technologies (ICGT), Trivandrum, 2012, pp. 041-045, doi: 10.1109/ICGT.2012.6477945.

[20] A. Inamdar, A. Sahu, J. Ren, S. Setoodeh, R. Mansour and D. Gupta, "Design and Evaluation of Flash ADC," in IEEE Transactions on Applied Superconductivity, vol. 25, no. 3, pp. 1-5, June 2015, Art no. 1400205, doi: 10.1109/TASC.2014.2365717.Young-Jae Cho and Seung-Hoon Lee, "An 11b 70-MHz 1.2-mm/sup 2/ 49-mW 0.18-/spl mu/m CMOS ADC with on-chip current/voltage references," in IEEE Transactions on Circuits and Systems I: Regular Papers, vol. 52, no. 10, pp. 1989-1995, Oct. 2005, doi: 10.1109/TCSI.2005.853251.

[21] F. Siqiang et al., "Enhanced offset averaging technique for flash ADC design," in Tsinghua Science and Technology, vol. 16, no. 3, pp. 285-289, June 2011, doi: 10.1016/S1007-0214(11)70041-4.

[22] Jyoti Sharma, Pallavi Bansal, Ayushi and Dr. Veepsa Bhatia, "Design of 3-bit Current Mode Flash ADC", 3rd IEEE International Conference on Recent Trends in Electronics, Information & Communication Technology (RTEICT-2018), pp. 652-657, 2018.

[23] K.Sai Kumar, K.Lokesh Krishna, K. Sampath Raghavendra and K.Harish, "A High Speed Flash Analog to Digital Converter", 2nd International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud) (I-SMAC)I-SMAC (IoT in Social, Mobile, Analytics and Cloud) (I-SMAC), pp. 283-288, 2018, DOI: 10.1109/I-SMAC.2018.8653782.

[24] G. Tretter, M. M. Khafaji, D. Fritsche, C. Carta and F. Ellinger, "Design and Characterization of a 3-bit 24-GS/s Flash ADC in 28-nm Low-Power Digital CMOS," in IEEE Transactions on Microwave Theory and Techniques, vol. 64, no. 4, pp. 1143-1152, April 2016, doi: 10.1109/TMTT.2016.2529599.

[25] S. Sheikhaei, S. Mirabbasi and A. Ivanov, "A 0.35 /spl mu/m CMOS comparator circuit for high-speed ADC applications," 2005 IEEE International Symposium on Circuits and Systems, 2005, pp. 6134-6137 Vol. 6, doi: 10.1109/ISCAS.2005.1466040.

[26] K. Sujatha, T. N. Bhagirath and K. K. Garje, "Design and simulation of high speed comparator for LVDS receiver application," 2015 Annual IEEE India Conference (INDICON), 2015, pp. 1-6, doi: 10.1109/INDICON.2015.7443604.

[27] Shashidhar.K, Chaithanya Mannepalli, Sreenivasa Rao Ijjada "1.5mW, 14.68V/µS- Low Power and high speed comparator design for ADC applications" International Journal of Innovative Technology and Exploring Engineering (IJITEE), vol. 8, issue 6S4, pp. 1322-1326,2019, DOI. 10.35940/ijitee.f1268.0486s419